

Homework 2 - Advanced Converter Topologies

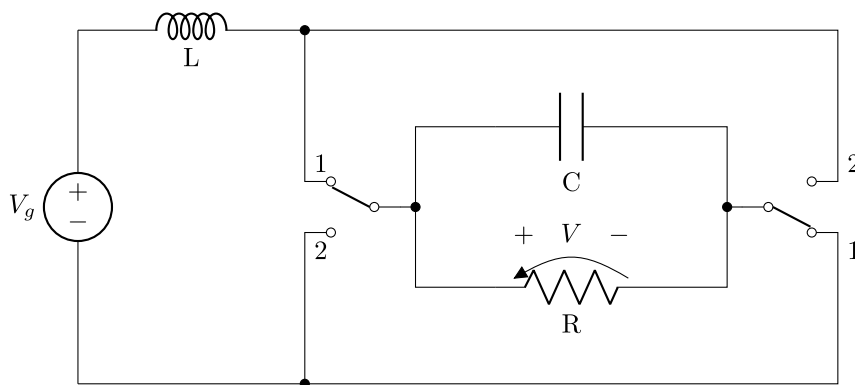
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1 Problem 1:



- The inductor has ohmic resistance R_L .
- All the other losses are neglected.
- The switches operate in a complementary manner, such that:
 - The switches on mode 1 in time interval $[0, DT_s]$.
 - The switches on mode 2 in time interval $[DT_s, T_s]$.

1.1 Question A:

The voltage amplification of the converter V/V_g : I will label the average current flows through the inductor as I , when the switches are in mode 1, the voltage across the inductor is:

$$V_L = V_g - V - V_L = V_g - V - IR_L$$

When the switches are in mode 2, the voltage across the inductor is:

$$V_L = V_g - (-V) - V_L = V_g + V - IR_L$$

Since the average voltage across the inductor over a full switching period is zero, I can write:

$$\begin{aligned} D(V_g - V - IR_L) + D'(V_g + V - IR_L) &= 0 \\ D(V_g - V - IR_L) + (1 - D)(V_g + V - IR_L) &= 0 \\ D(V_g - V - IR_L) - D(V_g + V - IR_L) + V_g + V - IR_L &= 0 \\ D(V_g - V - IR_L) + D(-V_g - V - IR_L) + V_g + V - IR_L &= 0 \\ -2DV + V_g + V - IR_L &= 0 \\ -2D + 1 = -2D + D' + D = D' - D \rightarrow V_g + (D' - D)V - IR_L &= 0 \end{aligned} \quad (1)$$

When the switches change their state, the current through the capacitor changes its direction. Using KCL, I can express the average current through the capacitor at each mode, as follows:
In mode 1:

$$I_{C_{\text{mode } 1}} = I - I_R = I - \frac{V}{R}$$

In mode 2:

$$I_{C_{\text{mode } 2}} = -I - I_R = -I - \frac{V}{R}$$

Because the average current through the capacitor over a full switching period is zero, I can write:

$$\begin{aligned} D \cdot I_{C_{\text{mode } 1}} + D' \cdot I_{C_{\text{mode } 2}} &= 0 \\ D \left(I - \frac{V}{R} \right) + D' \left(-I - \frac{V}{R} \right) &= 0 \\ D \left(I - \frac{V}{R} \right) + (1 - D) \left(-I - \frac{V}{R} \right) &= 0 \\ D \left(I - \frac{V}{R} \right) - D \left(-I - \frac{V}{R} \right) - I - \frac{V}{R} &= 0 \\ D \left(I - \frac{V}{R} \right) + D \left(I + \frac{V}{R} \right) - I - \frac{V}{R} &= 0 \\ DI + DI - I - \frac{V}{R} &= 0 \\ DI - (1 - D)I &= \frac{V}{R} \\ I &= \frac{V}{R(D - D')} \end{aligned} \quad (2)$$

Now, I can substitute equation (2) into equation (1), to find expression for V/V_g :

$$V_g + (D' - D)V - R_L \cdot \frac{V}{R(D - D')} = 0$$

$$V_g = R_L \cdot \frac{V}{R(D - D')} - (D' - D)V$$

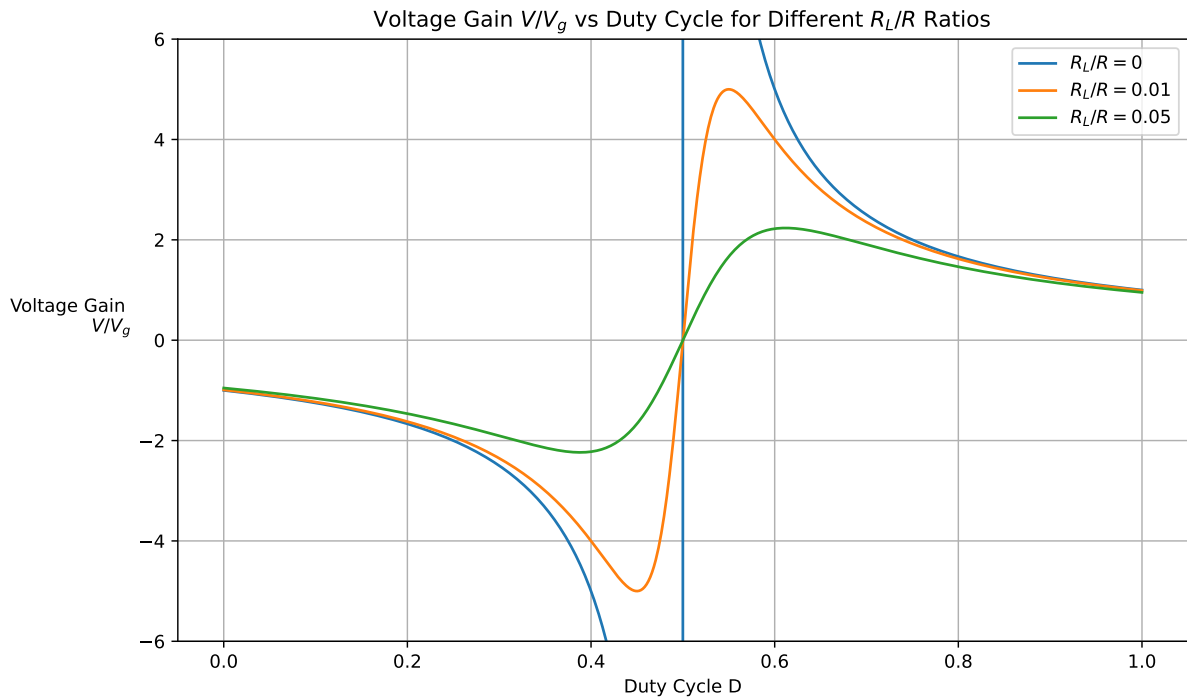
$$V_g = R_L \cdot \frac{V}{R(D - D')} + (D - D')V$$

$$V_g = V \cdot (D - D') \cdot \left(1 + \frac{R_L}{R(D - D')^2} \right)$$

$$\frac{V}{V_g} = \frac{1}{(D - D')} \cdot \frac{1}{1 + \frac{R_L}{R} \cdot \frac{1}{(D - D')^2}}$$

$$\frac{V}{V_g} = \frac{1}{2D - 1} \cdot \frac{1}{1 + \frac{R_L}{R} \cdot \frac{1}{(2D - 1)^2}}$$

1.2 Question B:



1.3 Question C:

To calculate the efficiency of the converter, I will compute the ratio between the output power and the input power:

$$\eta = \frac{P_{out}}{P_{in}}$$

~~This converter is not isolated, therefore, the current flowing into the converter from the source is equal to the current flowing out of the converter to the load.~~ because of KCL there is current through the capacitor! .

The input power can be expressed as:

$$P_{in} = V_g \cdot I$$

Since the current flowing through the resistor is not constant, I will express the output power as:

$$P_{out} = \frac{V^2}{R}$$

Now, I can express the efficiency as:

$$\begin{aligned} \eta &= \frac{P_{out}}{P_{in}} = \frac{V^2}{R} \cdot \frac{1}{V_g \cdot I} = \frac{V}{RI} \cdot \frac{V}{V_g} \\ \eta &= \frac{V}{R \cdot \frac{V}{(D-D')}} \cdot \frac{1}{(D-D')} \cdot \frac{1}{1 + \frac{R_L}{R} \cdot \frac{1}{(D-D')^2}} \\ \eta &= \frac{1}{1 + \frac{R_L}{R} \cdot \frac{1}{(D-D')^2}} \end{aligned}$$

1.4 Question D:

The equivalent circuit of the converter can be represented as follows: there would be a transformer that separates the input and output sides. On the input side, there would be a voltage source V_g in series with resistor R_L to represent the ohmic losses of the inductor. On the output side there would be the load resistor R . The main difficulty is to find the turns ratio of the transformer. In the answer to question A, equation (1) shows that:

$$V_g + (D' - D)V - IR_L = 0 \leftrightarrow V_g - IR_L = (D - D')V$$

Because of KVL, the right side of this equation is equal to the voltage drop over the input side of the transformer in the equivalent circuit. Thus, if on the output side there is voltage V , then the turns ratio of the transformer needs to be: $(D - D') : 1$. In this case, the input side of the transformer behaves as a dependent voltage source with ratio $(D - D') : 1$. On the other hand, equation (2) shows that:

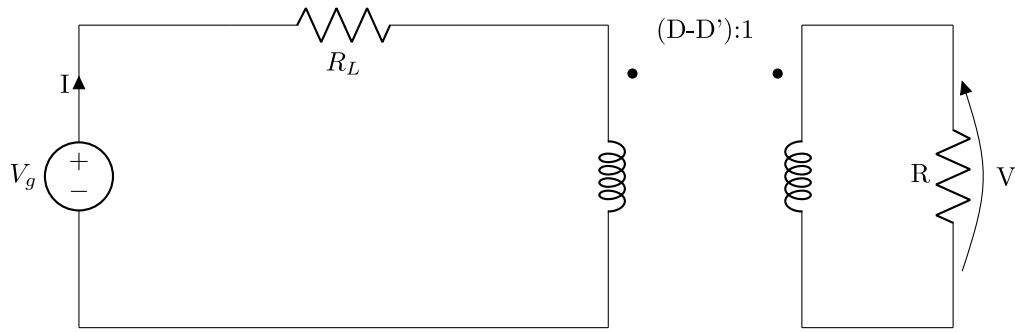
$$I = \frac{V}{R(D - D')} \leftrightarrow I \cdot (D - D') = \frac{V}{R}$$

I is the current on the input side, but V and R represent the values on the output side. Therefore, I can label $\frac{V}{R} = I_{load}$ and I get this ratio of currents between the input and the output side:

$$I \cdot (D - D') = I_{load}$$

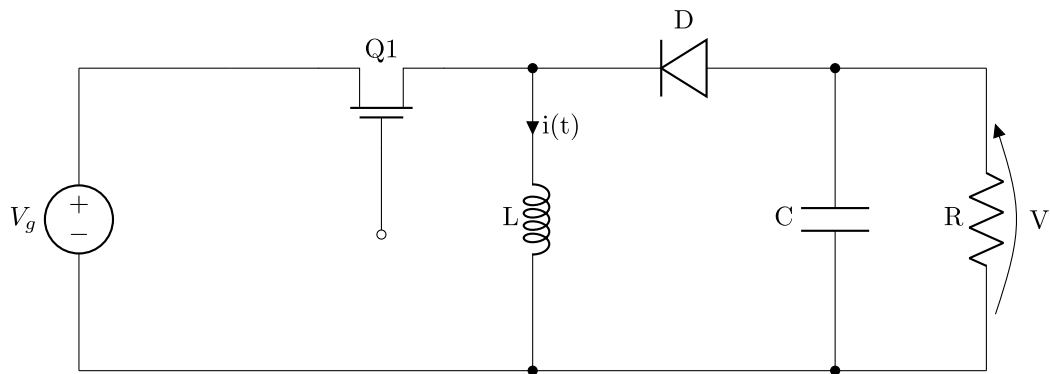
This means that the output side of the transformer in the equivalent circuit is a dependent current source with ratio $(D - D') : 1$.

Now, since I have the same ratio value for both the dependent voltage source on the input side and the dependent current source on the output side, the transformer turns ratio must be: $(D - D') : 1$.



2 Problem 2:

The following circuit is converter type BUCK-BOOST running in DCM mode.



- All the elements are ideal.
- All the losses are neglected.
- The transistor is conduction at time interval $[0, D_1 T_s]$.
- DCM mode.

2.1 Question A:

What is the condition for the converter to operate in DCM mode?

i The work principles of the converter for DCM mode are as follows:

- The voltages and currents in the system are periodic waves with period T_s , as result of that:
 - The average voltage across the inductor in one period is zero.
 - The average current through the capacitor in one period is zero.
- The following connections between the temporal waveforms and their average over period for all waves (voltages and currents):

$$\int_0^{T_s} x(\tau) d\tau = T_s \cdot X = \int_0^{T_s} X d\tau$$

in addition to that, for capacitor voltages (assuming small ripple):

$$\begin{aligned}\int_0^{D_1 T_s} v(\tau) d\tau &\approx D_1 T_s \cdot V \\ \int_{D_1 T_s}^{(D_1+D_2)T_s} v(\tau) d\tau &\approx D_2 T_s \cdot V \\ \int_{D_2 T_s}^{(D_2+D_3)T_s} v(\tau) d\tau &\approx D_3 T_s \cdot V\end{aligned}$$

First, I'll analyze the converter as it was operating in CCM mode, which will results into having equation for the average current through the inductor, I , and the equation of the current ripple through the inductor, Δi , having those two equations will allow me the derive the condition for operation in DCM mode by solving the inequality: $I < \Delta I \Rightarrow$ DCM mode.

2.1.1 phase 1, $0 < t < D_1 T_s$:

The switch is ON, the inductor is connected directly to the voltage source:

$$v_L(t) = V_g$$

The diode is reverse biased because the voltage source is connected to the cathode while the output voltage, which in buck boost is negative is connected to the diode's anode, therefor, there is no current loop between the capacitor and the voltage source, but also, **there is no current loop between the capacitor and the inductor**. however, in CCM, the output voltage across the load resistor is constant (Small-Ripple Voltage approximation), and applying KCL on the output node (negative voltage) result into current loop between the load resistor and the capacitor, with the sign convention where positive capacitor current flows from the output node to ground:

$$i_C(t) = -\frac{v(t)}{R} \approx -\frac{V}{R}$$

2.1.2 phase 2, $D_1 T_s \leq t < T_s$:

The switch is OFF, the inductor is now connected to the voltage output (diode voltage drop assumed to be neglect):

$$v_L(t) = v(t) \approx V$$

because current must be continues, the current which flowed through the inductor, $i_L(t)$ maintain it's direction (top to bottom), this results to current loop (counter-clockwise) between the inductor and the capacitor, applying KCL at the output node, and summarize the total currents leaving that node, I get that:

$$i_C(t) = -i_L(t) - \frac{v(t)}{R} \approx -I - \frac{V}{R}$$

2.1.3 Inductor Volt-Second Balance

The average voltage across the inductor in one period is zero:

$$\begin{aligned}
 0 &= \int_0^{T_s} v_L(t) d\tau = \int_0^{D_1 T_s} v_L(t) d\tau + \int_{D_1 T_s}^{T_s} v_L(t) d\tau \\
 &= V_g D_1 T_s + V D_1' T_s \Leftrightarrow V_g D_1 + V D_1' = 0 \\
 &\Rightarrow \boxed{V = -\frac{D_1}{D_1'} V_g} \tag{1}
 \end{aligned}$$

Reminder:

V is the average output voltage, and V_g is the source voltage.

2.1.4 Capacitor Charge (Amp-Second) Balance

The average current through the capacitor in one period is zero:

$$\begin{aligned}
 0 &= \int_0^{T_s} i_C(t) d\tau = \int_0^{D_1 T_s} i_C(t) d\tau + \int_{D_1 T_s}^{T_s} i_C(t) d\tau \\
 &= -\frac{V}{R} D_1 T_s + \left(-I - \frac{V}{R}\right) D_1' T_s \\
 \Leftrightarrow 0 &= -\frac{V}{R} D_1 + \left(-I - \frac{V}{R}\right) D_1' = -\frac{V}{R} D_1 - I D_1' - \frac{V}{R} D_1' \\
 &= -I D_1' - \frac{V}{R} (D_1 + D_1') = -I D_1' - \frac{V}{R} \\
 &\Rightarrow \boxed{I = -\frac{V}{R D_1'}} \tag{2}
 \end{aligned}$$

Reminder:

I is the average inductor current.

2.1.5 Inductor Current Ripple in CCM

The inductor current ripple, denoted as Δi (where the total peak-to-peak ripple is strictly defined as $2\Delta i$), is essential to find the boundary between CCM and DCM. To find the current ripple I will analyze the rate of change of the inductor current, from Farady's law of induction, the rate of change (aka slope) of the inductor current is proportional to the applied voltage:

$$\frac{di_L(t)}{dt} = \frac{v_L(t)}{L}$$

In phase 1, $v_L(t) = V_g$. therefore, the slope of the inductor current is constant and positive:

$$\frac{di_L(t)}{dt} = \frac{V_g}{L}$$

because $i_L(D_1 T_s) - i_L(0) = I + \Delta i - (I - \Delta i) = 2\Delta i$, I can substitute this into a slop line equation and get Δi :

$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} = \frac{\text{vertical change}}{\text{horizontal change}} \\ &\rightarrow \frac{V_g}{L} = \frac{2\Delta i}{D_1 T_s} \\ &\Rightarrow \boxed{\Delta i = \frac{V_g D_1 T_s}{2L}} \end{aligned} \tag{3}$$

2.1.6 DCM mode condition

Reminder:

$$I < \Delta I \Rightarrow \text{DCM mode}$$

first I'll plug in I and ΔI I found on equations 2 and 3:

$$I = -\frac{V}{RD'_1} < \Delta i = \frac{V_g D_1 T_s}{2L}$$

In the left hand side the I can express the output voltage, V , as function of the input voltage, V_g , by plug in Equation 1, this will result the following Inequality:

$$I = -\frac{V}{RD'_1} = -\frac{-\frac{D_1}{D'_1} V_g}{RD'_1} = \frac{D_1 V_g}{R (D'_1)^2} < \frac{V_g D_1 T_s}{2L}$$

reordering that inequality:

$$\begin{aligned} \frac{\cancel{D_1} V_g}{R (D'_1)^2} &< \frac{V_g \cancel{D_1} T_s}{2L} \\ \frac{1}{R (D'_1)^2} &< \frac{T_s}{2L} \\ \frac{2L}{RT_s} &< (D'_1)^2 = (1 - D_1)^2 \end{aligned}$$

1. on the left side I got "constant", K , that depend on L , R , T_s

$$K = \frac{2L}{RT_s} .$$

2. on the right side I got function K_{crit} whose argument is D_1 :

$$K_{crit} = (1 - D_1)^2 .$$

and, the converter operates in DCM when $K < K_{crit}$. $\square$